

BURNING SIGNALS, FORECASTING WILDFIRES

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Abstract: Wildfires are significant environmental hazards exacerbated by drought, low humidity, and climate change. Each year, they result in substantial economic losses globally and degrade air, soil, and water quality, which adversely affects the health of humans, animals, and plants. Large-scale fires destroy vegetation, displace wildlife, and disrupt ecosystems, creating an urgent need for effective detection and monitoring. This research presents a lightweight, low-computational-cost deep learning model based on a proxy normalized EfficientNet-b0, designed for wildfire detection. The proposed model classifies images into two categories: wildfire and no wildfire, using publicly available satellite images. Extensive experiments demonstrate that our model achieves an impressive 94% accuracy in detecting possible wildfires, with around 4 million trainable parameters. Compared to existing results, our architecture shows remarkable performance in wildfire detection.

Keywords: Proxy Normalization; EfficientNet; Squeeze-and-Excitation; Convolutional Neural Network; CNN; Wildfire; Lightweight Model; Deep Learning; Machine Learning; Wildfire Detection; Forecasting Wildfires

1. Introduction

According to Cambridge dictionary, wildfire is a fire that is burning strongly and out of control on an area of grass or bushes in the countryside [1]. In other words, Wildfire is an uncontrolled fire in an area of combustible vegetation like bush, desert, grass, peat, hill, or veld. It is mostly caused by accidental or intentional ignition, heatwaves, climate change, and lack of humidity. Climatic cycles with wet periods that create substantial fuels, followed by drought and heat, often play a significant role in severe wildfires. At a global level, human practices like industrialization, urbanization, and deforestation have impacted wildfire through climate change. The carbon released from wildfires can add to carbon dioxide in the atmosphere and henceforth contribute to the greenhouse effect which creates climate change feedback. Wildfires are destructive and recurring natural hazards which affects ecosystems, economies, and human health worldwide. They occur in areas like Mediterranean shrublands, boreal forests, and savannas, and their intensity has increased in recent decades. Wildfires spread quickly over vast areas, leaving behind long-term environmental and societal impacts. These include vegetation, economic, and biodiversity loss, and degradation of soil, water, and air quality. According to the National Oceanic and Atmospheric Administration (NOAA), fire seasons are becoming longer, less predictable, and more destructive, so there is an urgent need for reliable detection and monitoring solutions [2].

The consequences of wildfires is not just forest destruction. Economically, damages from major events often cross billions of dollars. These losses include homes, businesses, infrastructures, reduced agriculture, smoke-related illnesses, and reduced tourism. From

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January 7 to 31, 2025, the Los Angeles Metropolitan Area and San Diego County in California faced devastating wildfires, and by April 3, 2025, the wildfire had killed at least 30 people, forced more than 200,000 to evacuate, and destroyed over 18,000 homes. The wildfire burned more than 57,000 acres of land in total and resulted in property losses between \$76 billion and \$131 billion [2]. Environmentally, fires increase the rate of carbon emissions, releasing greenhouse gases such as carbon dioxide and methane which intensifies global warming. Public health impacts are significant, as wildfire smoke is strongly linked to asthma, cardiovascular diseases, and other respiratory conditions. Furthermore, exposure to large-scale fires imposes lasting psychological and social stresses on nearby communities.

Given the harmful impacts, researchers have recognized the importance of improving wildfire prevention, detection, and response systems. Traditional detection methods, such as human patrols, fire watchtowers, and citizen reporting, suffer from delayed response times and limited geographic coverage. Ground-based sensors and thermal cameras provide better local monitoring but are costly to deploy on a wide scale. Unmanned aerial vehicles (UAVs) are increasingly used in fire management, however their operational range is still restricted, and they require continuous human oversight. Remote sensing via satellite imagery has emerged as a key resource in wildfire detection and monitoring. Satellite sensors such as MODIS (Moderate Resolution Imaging Spectroradiometer) and VIIRS (Visible Infrared Imaging Radiometer Suite) have enabled global fire mapping and long-term monitoring. However, limitations in resolution, temporal frequency, and near real-time availability hinder their effectiveness for rapid detection and early intervention.

The rise of artificial intelligence has introduced new opportunities for using satellite imagery in wildfire detection. Convolutional neural networks (CNNs) have shown strong performance in image classification and pattern recognition tasks like natural disaster monitoring. CNN is a type of feedforward neural network that learns features via kernel optimization [3]. Several studies have applied CNN-based models like MobileNet, GoogleNet, ResNet, etc., to detect wildfires from optical or infrared satellite imagery. However, these approaches, especially deep CNN architectures, require tens of millions of trainable parameters, needing high computational power, extensive memory, and long training times. While they are effective in controlled settings, these models are unsuitable for real-time or embedded applications, as not everyone will have access to the high amount of resources required. This gap highlights the need for accurate yet lightweight models that balance performance with efficiency.

To address these challenges, this study introduces an enhanced architecture for wildfire detection based on a lightweight CNN known as EfficientNet-B0. The model is made up of several optimizations to improve parameter efficiency while maintaining high detection and prediction accuracy. First, a Proxy Normalized Activation function replaces traditional batch normalization. The difference is that batch normalization depends on mini-batch, whereas ProxyNormReLU normalizes features independently for each channel at each spatial location which makes training stable across varying batch sizes [4]. Second, a grouped convolutional layer reduces the computational cost by splitting channels into smaller groups and preserving feature maps without needing connectivity of full channel. Thirdly, a Squeeze-and-Excitation (SE) block is integrated to adjust channel responses and strengthen the model's ability to emphasize fire-relevant features. Altogether, these modifications form an EfficientNet-Enhanced architecture with approximately four million trainable parameters and achieves a detection accuracy of 94% on satellite images [5].

The choice of this model design reflects realistic deployment considerations. In many fire-prone regions, like underdeveloped countries, computational resources are limited, and connectivity is unreliable. Models designed for large data centers are not very useful

in those environments. A lightweight architecture can be embedded in mobile devices, drones, or edge-computing units, which expands the reach of wildfire monitoring to those areas. Efficient models also reduce energy consumption and training costs, supporting sustainable large-scale use. By focusing on parameter efficiency and portability, the proposed architecture aligns with the broader need for scalable and accessible wildfire management systems.

2. Literature Review

Many researchers studied wildfires and contributed to the analysis and prediction of wildfires. Categorizing the dataset into wildfire and no wildfire in many areas has received many attention. Various approaches like machine learning, and deep learning have been used to predict the wildfire. The use of AI for the prediction began to take shape in the 2010s and gained traction around 2015 when TensorFlow and satellite images supported machine learning. Toan et. al in 2019 [6] developed an autonomous wildfire detection system using continuous satellite imagery from GOES-16 and a deep learning architecture designed for pixel-level fire localization. Their model addresses key challenges in satellite data, such as temporal dependencies, spectral complexity, and cloud interference. The system achieved a 94% F1-score and 1.5× faster detection than baselines, with outputs visualized through an interactive dashboard for real-time analysis. This work highlights the effectiveness of AI-driven remote sensing for early wildfire detection and response. Another author Khryashchev et. al. in 2020 [7] proposed a convolutional neural network-based method for automated wildfire detection using high-resolution aerial and satellite RGB images. Using two datasets (1457 and 393 images), the authors trained a U-Net model with a ResNet34 encoder, applying data augmentation and windowing to enhance training. The model was optimized using the Adam algorithm and evaluated with metrics such as Dice coefficient, precision, recall, F1-score, and IoU. The results demonstrate the model's effectiveness for early wildfire detection in practical scenarios. Recently in 2025, Singh et. al. [8] explored the use of satellite imagery and Support Vector Machines (SVMs) for early wildfire detection in the Wolgan Valley, Eastern Australia. Using Landsat-8 data, including SWIR and NIR bands, and introducing the Normalized Difference Fire Index (NDFI), the study effectively detected active fires and classified burned areas. The SVM model demonstrated adaptability and high accuracy in identifying thermal anomalies and smoke plumes. While challenges remain—such as data availability and model interpretability—the study highlights the potential of combining spectral analysis and machine learning to enhance wildfire monitoring and response in fire-prone regions. In 2023, James et. al. [9] developed a lightweight wildfire detection system using satellite imagery and an optimized MobileNetV2 CNN. The model was trained with augmented data to handle seasonal variations and was designed for low-power devices like the Arduino Nano 33 BLE. Hyperparameter tuning showed that image size and network depth significantly affected performance and computational cost. The final model achieved strong classification accuracy while being 70% more efficient than a full-depth network, enabling real-time wildfire detection without ground control dependency. Rashkovetsky et. al. in 2021 [10] presented a deep learning-based workflow to detect wildfire-affected areas using satellite imagery across visible, infrared, and microwave domains. U-Net CNNs were trained separately on data from Sentinel-1, Sentinel-2, Sentinel-3, and MODIS sensors. The study found that combining Sentinel-2 and Sentinel-3 yielded the highest accuracy under clear conditions, while Sentinel-1 and Sentinel-2 fusion performed best under cloud cover. This highlights the effectiveness of multi-source data fusion and deep learning for accurate post-fire assessment. Prakash et. al. in 2023 [11] proposed an AI-based forest fire warning system using vision-based methods and deep learning for early detection. The approach

combines CNN transfer learning with Inception-v3 for classification, and introduces a hybrid RBFN-RAISR framework to enhance image resolution and accuracy. The model achieved high performance, with 97.55% accuracy and 93.33% F-score, outperforming earlier CNN methods. Additionally, a water wave optimization technique was used for feature selection and image enhancement, further improving fire detection reliability from visual data.

Li et al. (2024) developed FireNet, a multi-stage structure incorporating cross-channel attention, that identified fires in UAV imagery with 98.18% accuracy. Furthermore, the model showcased an average inference time of 10 milliseconds per image and operates at 86 frames per second (FPS) on embedded devices[12]. Bouguettaya et al. (2022) reviewed the use of UAVs with deep learning models such as YOLOv3 and DeepLabV3+, reporting F1-scores above 90% for fire segmentation [13]. Ghali et al. (2023) combined EfficientNet-B5 and DenseNet-201 with transformer models like TransUNet and TransFire, achieving F1-scores above 99%. These works inform the proposed model, emphasizing low computational cost and high precision [14]. Li et al. (2023) introduced SWVR, a lightweight model using RePViT and SimAM modules, to efficiently extract fire-related features with reduced computational complexity. It featured a bi-directional fusion network, incorporates lightweight Ghost Shuffle Convolution (GSConv), and uses the Wise Intersection over Union (WIoU) loss function. This lightweight model achieved 79.6% accuracy in detecting forest fires, which was a 5.9% improvement, and operated at 42.7 frames per second while reducing the model parameters by 11.8% and the computational cost by 36.5% [15]. The works done by researchers in the same field provide valuable foundations to this project to fine-tune parameters, reduce the number of parameters, and gain even higher accuracy

3. Background Works

3.1. Squeeze-and-Excitation Network

Squeeze-and-Excitation (SENet) is designed to improve CNNs by capturing channel interdependencies by minimizing computational overhead. It consists of two main operations: squeeze and excitation. The dimensions of input feature maps are reduced into a single numeric value during squeeze phase. It involves performing operations like average pooling over spatial dimensions of each channel, and introducing non-linearity using ReLU activation function. During excitation phase, the reduced features are restored back to the original size and sigmoid activation function is used to give output between 0 and 1. It produces the attention values that tells the model how much to emphasize or suppress each feature map. "Squeeze-and-Excitation" (SE) block adaptively recalibrates channel-wise feature responses by explicitly modelling interdependencies between channels. The blocks can be stacked together to form SENet architectures that generalise extremely effectively across different datasets. SE blocks bring significant improvements in performance for existing state-of-the-art CNNs at slight additional computational cost [16]. Overall, the following sequence forms the SE block:

1. Compress features.
2. Learn non-linear interactions.
3. Expand back to full channels.
4. Generate attention weights.

The figure 1 [17] shows the the block diagram of Squeeze-and-Excitation Network. It can be constructed for any transformation as ,

$$F_{tr} : X \rightarrow U, \quad X \in \mathbb{R}^{H' \times W' \times C'}, \quad U \in \mathbb{R}^{H \times W \times C} [18] \quad (1)$$

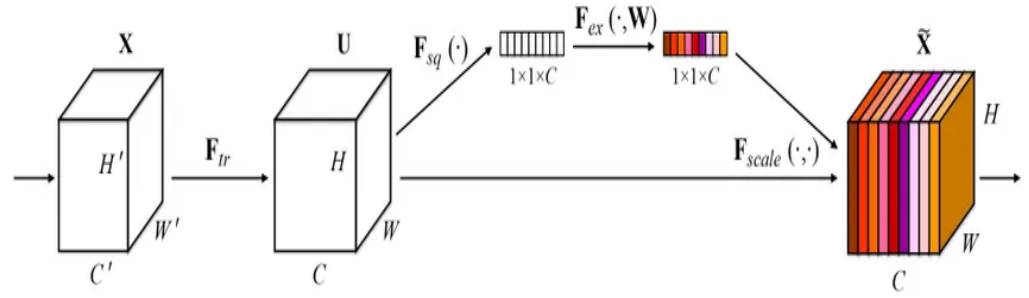


Figure 1. A block diagram for Squeeze-and-Excitation Network [17]

where F_{tr} is a convolutional operator. If $V = [v_1, v_2, \dots, v_C]$ is a set of learned filter channels, then the outputs of F_{tr} as $U = [u_1, u_2, \dots, u_C]$, where

$$u_c = v_c * X = \sum_{s=1}^{C'} v_c^s * x_s [18] \quad (2)$$

where $*$ represents convolution. v_c^s is a 2D spatial kernel. It acts as a single channel of v_c that operates on the corresponding channel of X . The output is the result of a summation across all channels that encompasses channel dependencies within v_c but the dependencies are intertwined with the spatial correlation captured by the filters [18].

3.2. EfficientNet

EfficientNet is a family of convolutional neural network which was published by the researchers at Google AI in 2019 for computer vision. One of its main advantages is that it has well balanced high accuracy rate and computational efficiency which is achieved through the method of compound scaling [19]. Compound scaling is a method of neural network architecture which scales the number of layers, number of channels in each layer and input image resolution of a neural network to improve its accuracy and reduce the computational costs. Instead of randomly scaling up, compound scaling is used to uniformly scale each dimension with a certain fixed set of scaling coefficients. Mathematically, it is achieved with the below equation [19]:

$$d = \alpha^\phi, w = \beta^\phi, r = \gamma^\phi \text{ such that } \alpha \cdot \beta^2 \cdot \gamma^2 = 2 \text{ where } \alpha \geq 1, \beta \geq 1, \gamma \geq 1$$

where d, w, r are depth, width and resolution respectively. The values of α, β, γ are determined using a grid search algorithm, and ϕ is called a compound coefficient determines the how much all three dimensions should be scaled simultaneously.

So, EfficientNet is fine-tuned to obtain high accuracy but there also are disadvantages if the network is very computationally heavy like network taking a lot of time to make predictions. This architecture uses a mobile inverted bottle neck convolution which is large due to the increase in FLOPS (Floating Point Operations Per Second). Compared to other architectures, EfficientNetB7 obtained around 84.4% and 97.3% accuracy on Imagenet, 91.7% on CIFAR-100 dataset, and 98.8% accuracy on the Flowers Dataset, with the model size being 8.4 times smaller and 6.1 times faster than the previous best CNN model.

3.3. Proxy Normalization

Proxy Normalization (PN) is a batch-independent [4] alternative to Batch Normalization (BN) that preserves BN's core effect by keeping pre-activations close to zero-mean and unit-variance without relying on batch statistics. PN first replaces the BN normalization

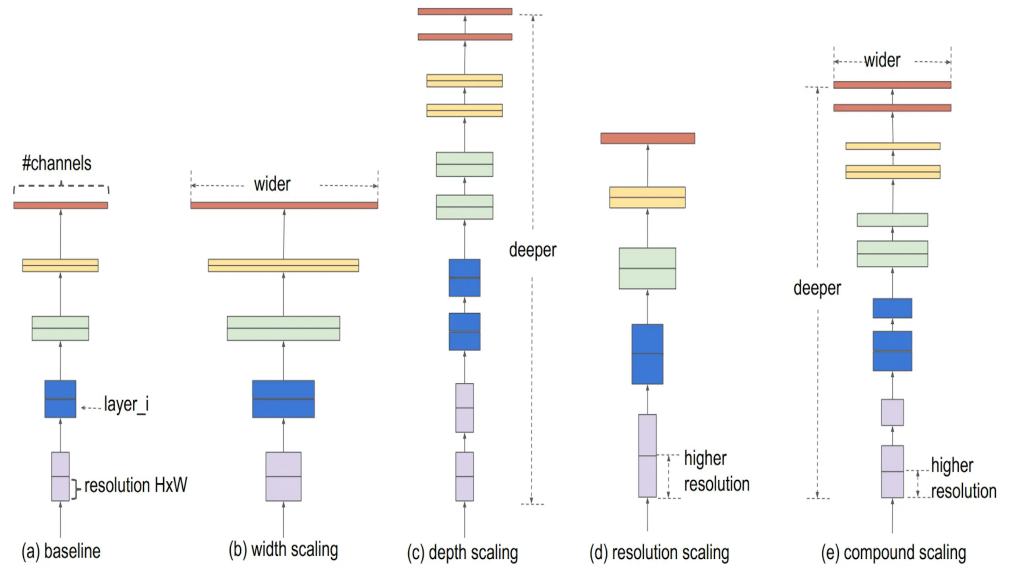


Figure 2. How compound scaling is used in EfficientNet

step with Layer Normalization (LN), which operates on each sample individually. Given an input tensor X , LN computes normalized pre-activations as shown in Equation 3 [4]:

$$Y_{b..} = \frac{X_{b..} - \mu_b}{\sqrt{\sigma_b^2 + \varepsilon}}, \quad (3)$$

where μ_b and σ_b are the mean and variance over all channels and spatial locations for each batch element b . After applying the affine transform and nonlinearity, PN introduces a proxy-normalization step that normalizes the activation using a learned Gaussian proxy distribution $\tilde{Y}_c \sim \mathcal{N}(\tilde{\beta}_c, (1 + \tilde{\gamma}_c)^2)$. The final normalized activation is computed as in Equation 4 [4]:

$$Z_{..c} = \frac{\phi(\gamma_c Y_{..c} + \beta_c) - \mathbb{E}_{\tilde{Y}_c}[\phi(\gamma_c \tilde{Y}_c + \beta_c)]}{\sqrt{\text{Var}_{\tilde{Y}_c}[\phi(\gamma_c \tilde{Y}_c + \beta_c)] + \tilde{\varepsilon}}}, \quad (4)$$

which encourages activations to remain close to normalized, effectively matching Batch Normalization performance while completely removing dependence on batch size.

4. Proposed Architecture

The proposed model is light-weight, spatial, and novel model which is designed for accurate detection of wildfire in a given region. The main objective of this model is reduce computational cost while also having high accuracy and lower computational time, so that it can be accessed in areas with lower available resources. The proposed model utilizes EfficientNet-B0 with Squeeze and Excitation layers with a specialized module to improve feature normalization and channel recalibration. The goal is to build a classifier which is capable of learning representations from image data. The design combines grouped convolutions, a proxy normalization and activation scheme, and a squeeze-and-excitation block before classification.

The main model is EfficientNet-B0 which is selected for its compound scaling of depth, width, and resolution. The pretrained network is reduced by removing the global pooling and classifier layers by using "nn.Identity()" which acts as a "no-op" operation. It is a way to effectively "remove" a layer or module without physically deleting it from the model's

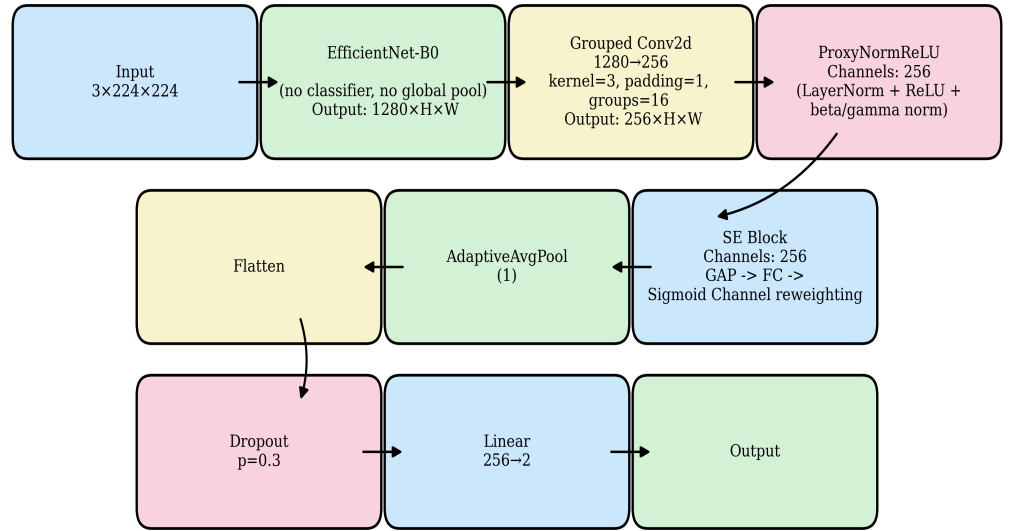


Figure 3. Proposed EfficientNet-Enhanced Architecture for Wildfire Detection

structure. This provides a feature map of dimension 1280 channels. Instead of passing these features directly into a fully connected head, a grouped convolution is applied, which reduces the number of channels to 256. Mathematically, for input tensor $X \in \mathbb{R}^{B \times C \times H \times W}$, the grouped convolution performs [19]:

$$Y = W_g * X + b \quad (5)$$

where g denotes the number of groups (16), and W represents convolution kernels shared within groups. The number of kernels used here is 3.

The reduced feature map is then normalized and activated through ProxyNormReLU which is a replacement for batch normalization. Regular batch normalization requires statistics across the mini-batch, which introduces instability in small-batch training. ProxyNormReLU instead applies layer normalization per pixel across channels which is then followed by a learnable reparameterization. For input vector $x \in \mathbb{R}^C$ [4],

$$\hat{x} = \frac{x - \mu}{\sigma}, \quad \mu = \frac{1}{C} \sum_{i=1}^C x_i, \quad \sigma = \sqrt{\frac{1}{C} \sum_{i=1}^C (x_i - \mu)^2}. \quad (6)$$

$$y = \frac{\text{ReLU}(\hat{x}) - \beta}{\gamma}, \quad (7)$$

where β and γ are learnable parameters and $\text{ReLU}(\hat{x}) = \max(0, \hat{x})$.

To enhance channel-wise attention, the network uses a squeeze-and-excitation (SE) block. The SE mechanism performs global average pooling across spatial dimensions, producing a descriptor $z \in \mathbb{R}^C$ [17]:

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W x_{c,i,j}. \quad (8)$$

The descriptor z is passed through a two-layer fully-connected network (with reduction ratio r) that produces channel weights $s \in [0, 1]^C$. The weights are broadcast back and applied elementwise to the feature maps:

$$\tilde{X}_c = s_c \odot X_c, \quad (9)$$

where $\odot$ denotes element-wise multiplication and $X_c \in \mathbb{R}^{H \times W}$ is the c -th channel feature map.

which emphasizes more informative channels and suppresses less useful ones.

The head of the model collects information through average pooling to produce a single vector per channel. This is followed by flattening, dropout for regularization, and a final linear classifier projecting from 256 dimensions to the number of classes. The forward flow is therefore:

Input $\rightarrow$ EfficientNet-B0 $\rightarrow$ Grouped Conv $\rightarrow$ ProxyNormReLU $\rightarrow$ SE Block
 $\rightarrow$ GlobalPooling $\rightarrow$ Dropout $\rightarrow$ Fully Connected Layer

This architecture addresses three limitations of baseline convolutional models. First, grouped convolution reduces the parameter count and computational cost while preserving local structure. Second, ProxyNormReLU stabilizes training without relying on batch, which is important for high-resolution inputs or small batch sizes. Third, the SE block introduces dynamic channel reweighting which enables the network to adaptively alter feature maps and improve class separability.

By combining an EfficientNet with these improvements, the proposed model balances accuracy and efficiency. It influences pretrained features, introduces stable normalization, and includes adaptive attention. Together, these modifications are expected to provide stronger generalization and more reliable predictions for binary or multi-class classification tasks.

5. Performance Evaluation

There are various metrics used to evaluate performance of a model in machine learning. Some of them are accuracy, precision, sensitivity, F1_Score, error, mean squared error(MSE), over segmentation rate(OR), under-segmentation rate(UR), Dice SImilarity coefficient(DSC), Jaccard Score(JS), etc. Among all these, we used accuracy, precision, recall, and F-1 Score metrics to evaluate how good our model is at achieving its goal of accurately predicting the wildfires. These metrics ensures that the proposed model is accurate for training data as well as new and unseen dataset. There are also some terms used like TP, TN, FP and FN. TP is True Positives and TN means True Negatives. TP is the number of correctly predicted positive instances by the model. For example, if there is possible wildfire in the certain area, the model predicts wildfire. TN is the number of negative instances correctly predicted by the model. This means that is there is no possible wildfire in the certain area, the test results are negative. Finally, FP stands for False Positives and FN stands for false Negatives. FP is the number of negative instances incorrectly predicted as positive by the given model. This means that the test can show that there is wildfire even if there are no wildfires. FN is the number of positive instances that the model incorrectly predicts as negative which means that in the area with wildfire, the model is going to predict that there is no possible wildfire. The metrics are described below:

5.1. Accuracy

Accuracy measures how correct the trained parameters are and gives output as the ratio of correct observations out of the total observations. The accuracy is calculated as [20],

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (10)$$

When TP and TN are high in comparison to the total predictions, the accuracy is high. On the other hand, if the numbers of TP and TN are low compared to the total predictions, the accuracy is low. The total prediction is the sum of all the predictions: TP + TN + false positives (FP) + false negatives (FN). So, we need high accuracy in order to enhance the reliability of model's performance.

5.2. Precision

Precision can be seen as the measure of quality and it is calculated as the ratio of True Positives to the sum of True Positives and False Positives. The equation to calculate the precision is defined as [20],

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

According to the equation, a high precision value means that when the model predicts a positive outcome, it is usually correct. This also suggests that the model has a low number of false positives which makes positive predictions reliable. However, a low precision value indicates that the positive predictions made by the model are incorrect. Therefore, we can conclude that low precision can have a negative impact.

5.3. Recall

Recall(sensitivity), unlike precision, is the measure of "quantity". It is the model's capability to accurately predict all the positives in a dataset. It is calculated as the ratio of True Positives to the sum of true positives and false negatives. The formula to calculate recall is as follows [20],

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

Based on the above equation, we can say that a high recall means that the model predicts most actual positives, while a low recall score indicates a high number of false negatives. A low recall score could result in significant issues and can also have negative impact to the overall reliability of the model.

5.4. F1-score

In the statistical analysis, the F1 score is used to measure predictive performance. It is calculated as the ratio of two times precision and recall to the sum of precision and recall. The F1 score can also be defined as the harmonic mean of precision and recall. Its values range from 0 to 1, where 0 indicates the lowest performance and 1 indicates the highest performance [20].

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (13)$$

Based on the equation above, a high F-1 score indicates that the model identifies a greater proportion of positive instances while minimizing false positives. Whereas, a low F-1 score suggests that the model struggles to accurately predict positive instances.

6. Experimental Results

6.1. Dataset

Table 1. Approximate distribution of wildfire and no-wildfire samples across the training, testing, and validation datasets.

Data Samples	Wildfire	No Wildfire	Total
Training Data Samples	730	1158	1888
Testing Data Samples	159	251	410
Validation Data Samples	157	245	402
Total	1046	1654	2700

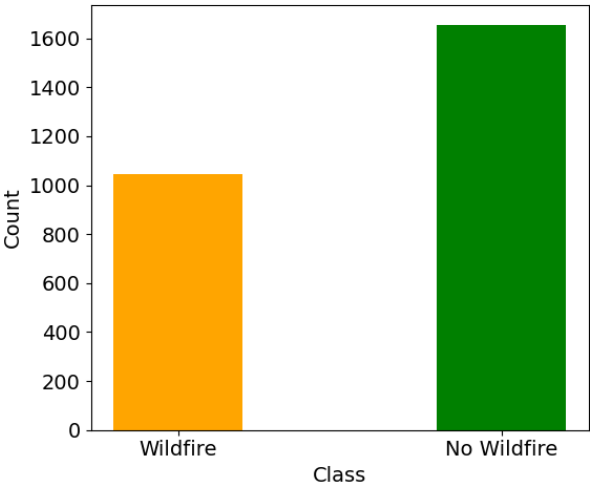


Figure 4. Class distribution of the wildfire dataset

This research uses the satellite image dataset from publicly available on Kaggle [5]. This dataset is divided into three main directories: test, train, and validation. Out of total of 2700 images, 70% of them is training dataset, 15% of them is testing dataset, and 15% of them is validation dataset. All of them are approximate division of the total dataset. Each of the directories is divided into two main categories: wildfire and no wildfire where wildfire is comprised of 39% and no wildfire is comprised of 61%.

The wildfire dataset symbolizes an effort to explore the capabilities of RGB imagery in the realm of forest fire detection through machine learning methodologies. The images were taken from platform like government databases, Flickr, and Unsplash emphasizing the dataset’s commitment to transparency. This dataset was prepared to bolster the model’s accuracy and reduce false alarms. Its merits become particularly evident when benchmarked against traditional classification techniques [5].

These images are augmented and preprocessed with techniques like resizing, horizontal flip, random rotation, color jitter, and normalization of each color. The images were originally 350 × 350 pixels, with some being more or less than the defined size. All of the images were resized into 160 × 160 size, randomly rotated during training for -15 to 15 degrees, set the constant brightness, contrast, and saturation to 0.2, and hue to 0.1, and normalized and converted raw pixel value to mean ‘0’ and standard deviation close to 1 in order to retain uniform size and type. All images are freely and publicly available to AI researchers, making them an invaluable resource for creating more effective diagnostic tools. The main dataset (Wildfire Prediction Dataset) was divided into batch size of 32 and shuffled while training. The distribution is summarized in Table 1.

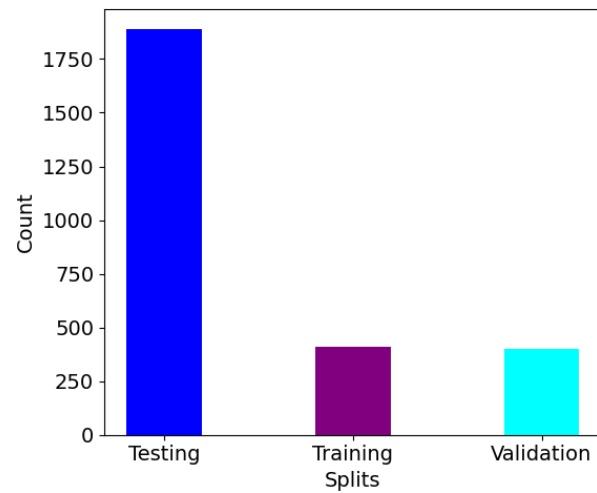
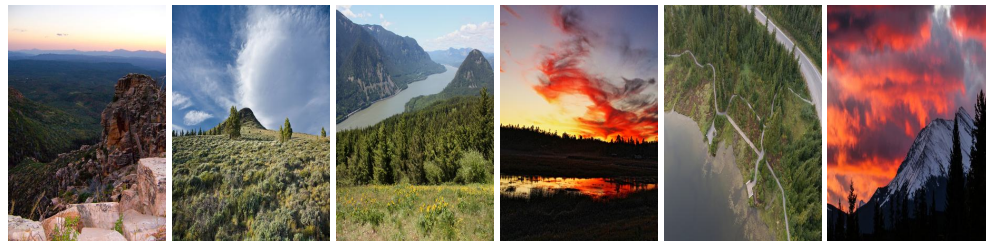


Figure 5. Test, Train, and Validation of the wildfire dataset.



(a) Sample Fire class images



(b) Sample NoFire class images

Figure 6. Sample images from the wildfire dataset

6.2. Methods

The dataset preprocessing consisted of the process of transforming all the images into same size so that it is easier for model training. We ran onto the problem of size overload before preprocessing because model was not able to identify some of the images in the dataset because of the size difference. These images are augmented and preprocessed with techniques like resizing, horizontal flip, random rotation, color jitter, and normalization of each color. The images were originally 350×350 pixels, with some being more or less than the defined size. All of the images were resized into 160×160 size, randomly rotated during training for -15 to 15 degrees, set the constant brightness, contrast, and saturation to 0.2, and hue to 0.1, and normalized and converted raw pixel value to mean 0 and standard deviation close to 1 in order to retain uniform size and type. All images are freely and publicly available to AI researchers, making them an invaluable resource for creating more effective diagnostic tools. The main dataset (Wildfire Prediction Dataset) [5] was divided into batch size of 32 and shuffled while training.

The model architecture was based on a convolutional neural network with an integrated Squeeze-and-Excitation (SE) block into main EfficientNet-B0 block to enhance channel-wise feature weighting, and proxynormalized for removal of batch dependence. The SE block consisted of a sequence of fully connected layers with ReLU and Sigmoid activations to compute adaptive feature importance. The implementation was carried out in Python using the PyTorch framework. The primary libraries included Torch for deep learning, sklearn.metrics for classification report and confusion matrix, matplotlib for graphical interpretation and visualization, seaborn for easing the use in data visualization, timm for image models, layers, utilizers, optimizers, etc, and OS module to let the users interact with the native OS python that is currently running on. The model was trained on GPU using CUDA acceleration for faster computation. The model uses ProxyNormReLU for normalization and activation which replaces batch normalization. The main purpose of this block is to provide stable activations without replying on batch statistics, improving performance on small or non stationary datasets. The squeeze and Excitation block improves channel wise feature attention to emphasize the important channels and suppress less relevant ones. EfficientNet-B0 is the main model architecture which consists of a group convolution to reduce channel dimensions from 1280 to 256, with 16 groups. The main process of forward pass is: Image $\rightarrow$ EfficientNet feature extraction $\rightarrow$ grouped convolution $\rightarrow$ ProxyNormReLU $\rightarrow$ SEBlock $\rightarrow$ pooling $\rightarrow$ classification. Combining EfficientNet, ProxyNormReLU, and SEBlock strengthens the model by improving feature extraction, normalization stability, and channel-level attention. Each component solves a specific weakness in typical convolutional neural networks, and together they create a more efficient and robust wildfire prediction system.

6.3. Setup

PyTorch deep learning framework was used to implement the proposed model. The model was trained, tested, and experiment on a workstation computer with an Intel(R) i9 13900K CPU, 128 GB memory, and NVIDIA RTX A6000 GPU(48 GB). The training was performed using Adam optimizer with patience of 2, factor of 0.5, learning rate of $1e-4$, and batch size of 32. The learning rate was reduced by a factor of 0.1 if the validation loss did not improve for 5 consecutive epochs. The model was trained for 20 epochs at first, and then slowly increased to 120 and then to 220 based on validation accuracy to prevent overfitting. Cross-entropy loss function was used for binary classification between wildfire and no wildfire classes. Data augmentation techniques such as random horizontal flips, rotations, and color jittering were applied during training. The final model weights were selected based on the best validation accuracy achieved during training and saving the best model and re-testing the dataset on that model.

6.4. Result Analysis

This section presents the results gained after training, testing, and validating more than 2,700 dataset separated into two main classes: wildfire and no wildfire. The model was first evaluated on validation data and used to hypertune the parameters and assess the model's performance on data it has not directly learned from, and later evaluated again on test data to determine overall performance. The results obtained from this project is astounding and is reported in the table 2.

Table 2. Performance metrics of the proposed EfficientNet-Enhanced model for wildfire prediction.

Metric	Value (%)
Accuracy	94.14
Precision	93.50
Recall	91.50
F1-Score	92.40
Existing Model Accuracy	91.22

The proposed model achieved an accuracy of 94.14%, precision of 93.5%, recall of 91.5%, and F1-score of 92.4%. These results indicate that the model is highly effective in distinguishing between wildfire and no wildfire instances. The high precision value suggests that when the model predicts a wildfire, it is usually correct, minimizing false positives.

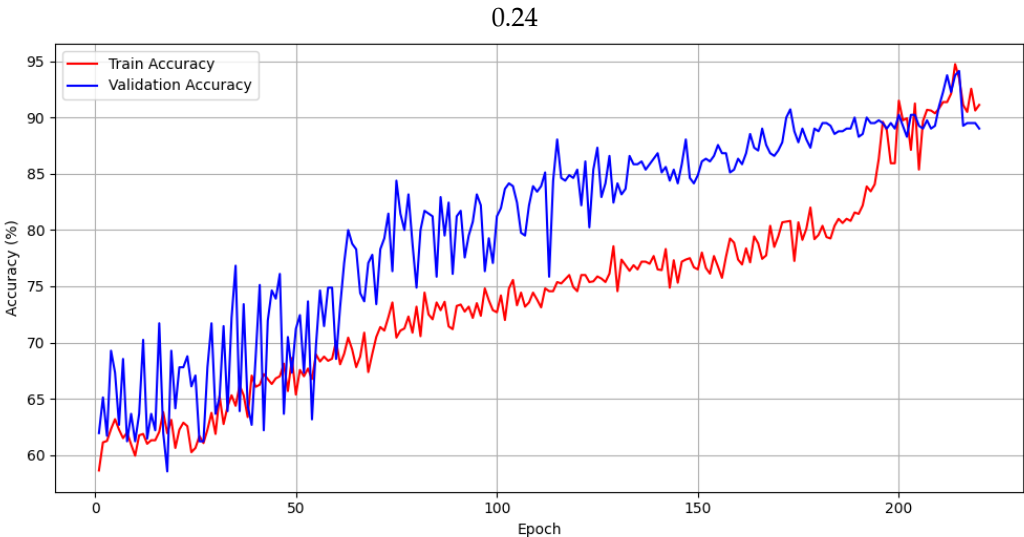


Figure 7. Epoch Vs Accuracy for training and validation

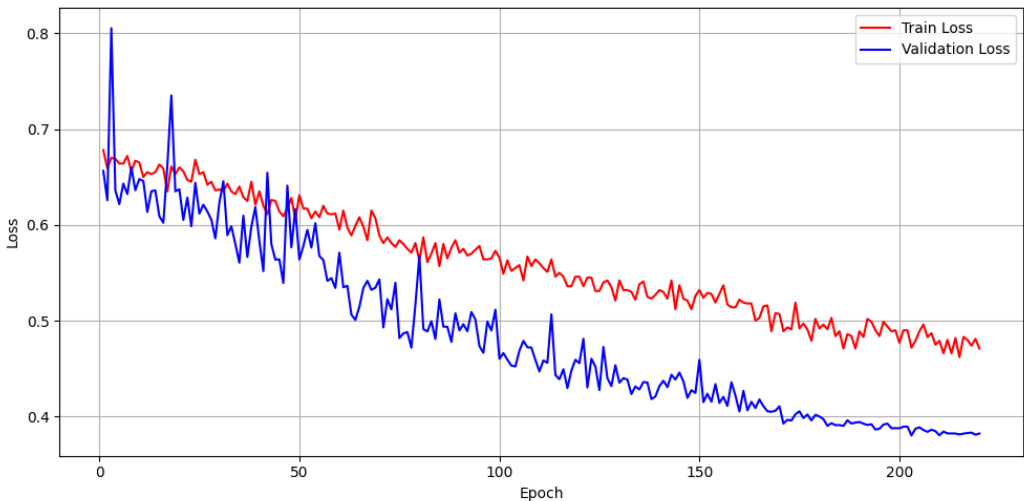


Figure 8. Loss Vs Epoch for training and validation

The training and validation accuracy and loss curves are shown in figures 7 and 8 respectively for 220 epochs, batch size 32, momentum 0.9, and weight decay of 5e-05. The training accuracy steadily over epochs which is indicating that the model is learning from the training data. The validation accuracy also improves which is suggesting good

generalization to unseen data. The loss curves show a decreasing trend for both training and validation which indicates that the model is minimizing the error effectively.

The confusion matrix is a tabular layout that allows unsupervised learning visualization, and is also known as error matrix or matching matrix. The rows represent the actual class and column represents the predicted class or vice versa. Therefore, the diagonal represents all the instances of correct predictions. The confusion matrix in figure 9 illustrates the model’s performance in classifying wildfire and no wildfire instances. The matrix 145 images were correctly identified (True Positives), while 14 were misclassified as no wildfire (False Negatives). For the no wildfire cases, 241 were accurately predicted (True Negatives), with 10 incorrectly labeled as wildfire (False Positives). This indicates a high level of accuracy in distinguishing between the two classes, with a low rate of misclassification. The model demonstrates strong predictive capabilities, effectively identifying wildfire occurrences while minimizing false alarms.

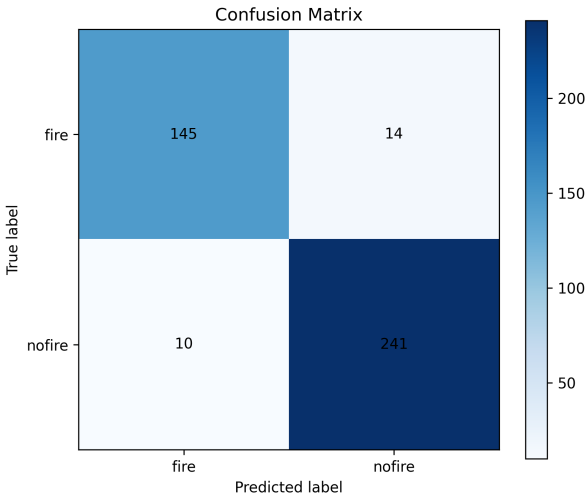


Figure 9. Confusion Matrix for Fire vs NoFire Classification

6.5. Result Comparison

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Table 3. Wildfire prediction literature summary and comparison.

Author (Year)	Dataset	Model Used	Evaluation Metrics
Rashkovetsy et al. (2021)	Cal Fire database	S2-S3 bi-instrument model	Precision 0.87, Recall 0.87, F1 0.87
Rashkovetsy et al. (2021)	Cal Fire database	S1-S2 bi-instrument model	Recall 0.92
Nguyen Thanh Toan et al. (2019)	GOES-16	Advanced deep learning architecture	F1 94%
James et al. (2023)	Sentinel-2	MobileNetV2 (50 percent depth reduction)	Accuracy 70%
Bahadori et al. (2023)	Landsat-8, MODIS	RNN	AUC 0.971 (MODIS), 0.966 (Landsat-8)
Bahadori et al. (2023)	Landsat-8, MODIS	LSTM	AUC 0.964 (MODIS), 0.941 (Landsat-8)
Sayad et al.	Databricks wildfire dataset	Remote sensing and data mining model	Precision 0.98
Subramanian and Crowley	Northern Alberta, Canada	Value Iteration (reinforcement learning)	Accuracy 79.6
Subramanian and Crowley	Northern Alberta, Canada	A3C deep RL model	Accuracy 87.3
Asanjan et al. (2023)	MODIS, VIIRS	Stochastic ReLU U-Net	Precision 0.60, Recall 0.35, F1 0.43, Jaccard 0.39
Asanjan et al. (2023)	MODIS, VIIRS	Dropout U-Net	Precision 0.53, Recall 0.30, F1 0.40, Jaccard 0.34
Asanjan et al. (2023)	MODIS, VIIRS	Probabilistic U-Net	Precision 0.56, Recall 0.57, F1 0.55, Jaccard 0.46
Phan Thanh Cong, Nguyen Thanh Tam	GOES-16	3D CNN	F1 0.94
Hoang et al.	GOES-16 (AWS-3)	Latent Residual Dynamic Model	Only reported loss graph
Ali and Kurnaz (2024)	Landsat, Sentinel-1, Sentinel-2	U-Net, CNN	Accuracy 82%
Existing EfficientNet-B0	Kaggle Dataset	EfficientNet	Accuracy 91.22%
Our Proposed Model	Kaggle Dataset	Enhanced EfficientNet	Accuracy 94.14%, Precision 93.5%, Recall 91.5%, F1 92.4%

We compared the proposed model other researchers' works in the same field. The comparison is summarized in the table 3. The proposed model achieved an accuracy of 94.14%, precision of 93.5%, recall of 91.5%, and F1-score of 92.4%. Compared to other models, such as Rashkovetsy et al. (2021) who achieved an F1-score of 87% using S2-S3 bi-instrument model, and James et al. (2023) who reported an accuracy of 70% with a MobileNetV2 architecture, the proposed model demonstrates superior performance across all evaluated metrics. Additionally, models like those by Asanjan et al. (2023) using various U-Net architectures reported lower precision and recall values, further highlighting the effectiveness of the proposed EfficientNet-Enhanced architecture for wildfire prediction tasks.

7. Conclusion and Future works

Lightweight deep learning models for wildfire prediction using satellite imagery are important for timely detection and response because not all of the regions can access non-lightweight models even though they are very prone to wildfires. This research proposed an EfficientNet-Enhanced model that assembles Squeeze-and-Excitation blocks and Proxy Normalization to improve accuracy while minimizing computational costs. The model achieved high performance metrics, including an accuracy of 94.14%, precision of 93.5%, recall of 91.5%, and F1-score of 92.4%, outperforming several existing approaches. The grouped convolutions and adaptive channel attention mechanisms contributed to the model's efficiency and effectiveness in distinguishing wildfire from non-wildfire instances. However, there are still some limitations to this model. The model's performance may be different across different geographic regions and seasons due to variations in vegetation, weather conditions, and fire behavior. Additionally, the model relies on high-quality satellite imagery, which may not always be available in real-time or for all areas. There is also a risk of false positives and false negatives, which could lead to unnecessary evacuations or missed fire detections.

Regardless, the model's interpretability could be improved to better understand the features driving predictions. Future work could explore the integration of temporal data from satellite image sequences to capture fire dynamics over time and improve prediction accuracy. There are 4 million parameters in the model currently which could be reduced further to make it more lightweight. Additionally, embedding the model into devices such as mobile phones, drones or sensors could help in real-time and quick wildfire detection in remote areas with limited or less computational resources.

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